



Marc Seguin's bridge across the Rhône between Tournon-sur-Rhône and Tain-l'Hermitage used a pioneering wire cable suspension system.

THE OPENING OF THE STOCKTON AND DARLINGTON railway on September 27 in the north of England marked the **beginning of the railroad age**. Smaller steam railways had existed before, but this project involved massive bridges and viaducts. The railway was engineered by **George Stephenson**, who also designed its first locomotive, *Locomotion No.1*.

One early passenger was French engineer **Marc Seguin** (1786–1875), whose experience inspired him to create his own steam railways in France. In August, Seguin also opened **Europe's first large wire cable suspension bridge** between Tournon-sur-Rhône and Tain-l'Hermitage, which spanned almost 298 ft (91 m).

Another technological first was an **electromagnet that was capable of supporting**

8 PERCENT THE AMOUNT OF ALUMINUM PRESENT IN EARTH'S CRUST

more than its own weight.

Made by British electrical engineer **William Sturgeon** (1783–1850), the 7 oz (200 g) magnet could lift 9 lb (4 kg).

The Danish physicist who discovered electromagnetism in 1820, **Hans Ørsted**, created **aluminium** in a chemical reaction in 1825.

British scientist **Michael Faraday**, who was another pioneer of electromagnetism, **discovered benzene** in the oil residue that was created from making coal gas for gaslights. Benzene is a key ingredient of petroleum and is used to make plastic.

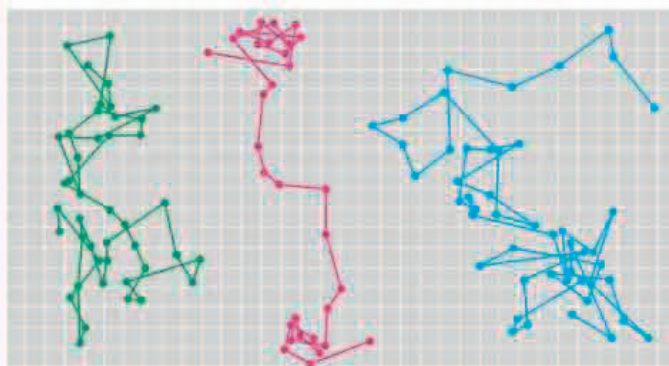
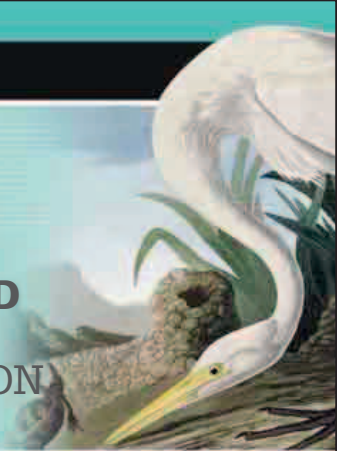
Meanwhile, French naturalist **Georges Cuvier** published his idea (first proposed in 1812) that large groups of animals had been wiped out by past catastrophes, in the book *Discourse on the Revolutions of the Surface of the Globe*. German geologist **Christian von Buch** (1774–1853) argued that natural variations between animals led to **separate species**.

Stockton and Darlington railway
The opening of George Stephenson's Stockton and Darlington Railway was a major event, attracting over 40,000 spectators and worldwide attention.



500

THE NUMBER OF **BIRD SPECIES** PAINTED BY **JOHN JAMES AUDUBON**



BROWNIAN MOTION

In 1827, when Robert Brown observed pollen grains in water through a microscope, he noticed they moved around at random, but could offer no explanation for it. In 1905 Albert Einstein (1879–1955) showed that the pollen grains were being knocked by water molecules, which caused them to move. This map of Brownian motion shows the random path of individual grains.

IN THE NATURAL WORLD, Russian naturalist **Karl von Baer** (1792–1876) discovered in 1826 that mammals start life as eggs or ova, and Scottish biologist **Robert Grant** (1793–1874), German naturalist **August Schweigger** (1782–1821), and German anatomist **Friedrich Tiedemann** (1781–1861) all argued that both **plants and animals have a common origin**.

Another Russian, mathematician **Nikolai Lobachevsky** (1792–1856), presented his system of **hyperbolic geometry** in

February 1826, involving imaginary surfaces and lines.

Two important developments in engineering also occurred in 1826. American inventor **Samuel Morey** (1762–1843) patented an **early version of the internal combustion engine**.

In the summer of 1826, French inventor **Joseph Nicéphore Niépce** (1765–1833) used a camera obscura to take the world's oldest-known **photograph** on a light-sensitive, bitumen-coated, pewter plate.

In nearby Montpellier, French chemist **Antoine Balard**

